

Firmware Upload Guide

Step 1. Install Arduino IDE

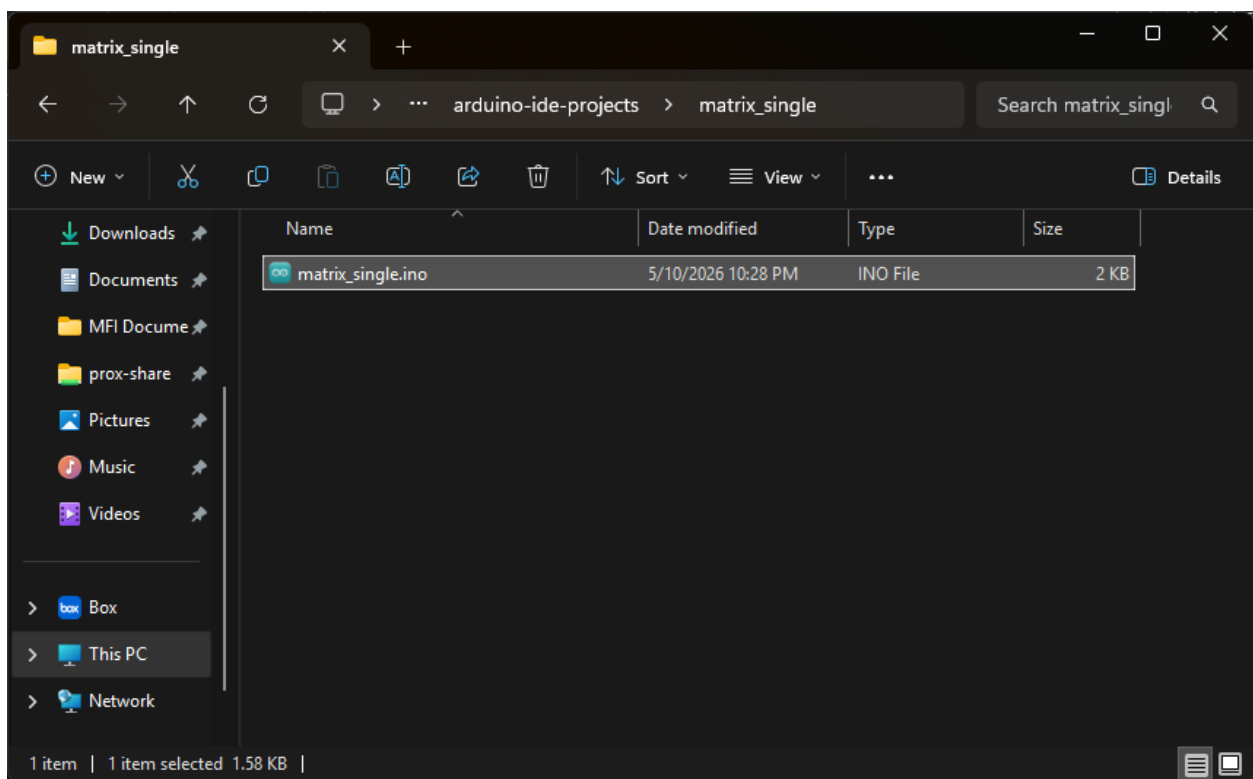
- Follow steps here: <https://www.arduino.cc/en/software/>

Step 2. Setup Arduino IDE for ESP 32

- Follow steps here: <https://developer.espressif.com/blog/2025/10/arduino-get-started/>

Step 3. Open a project

- Open .ino file in the projects

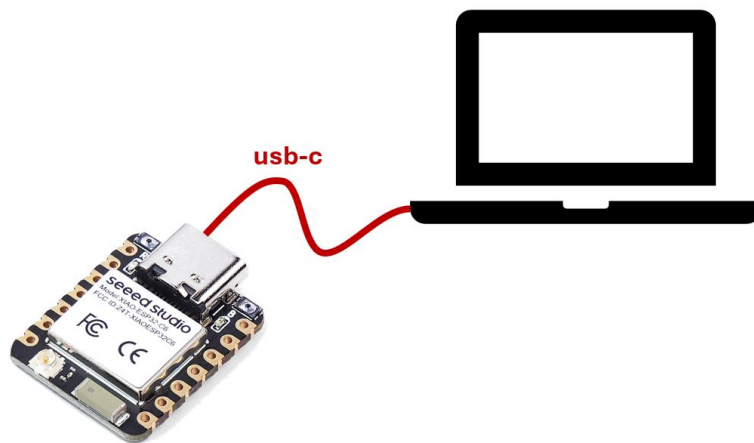


```
matrix_single | Arduino IDE 2.3.8
File Edit Sketch Tools Help
[Icons] Select Board [Dropdown] [Icons]

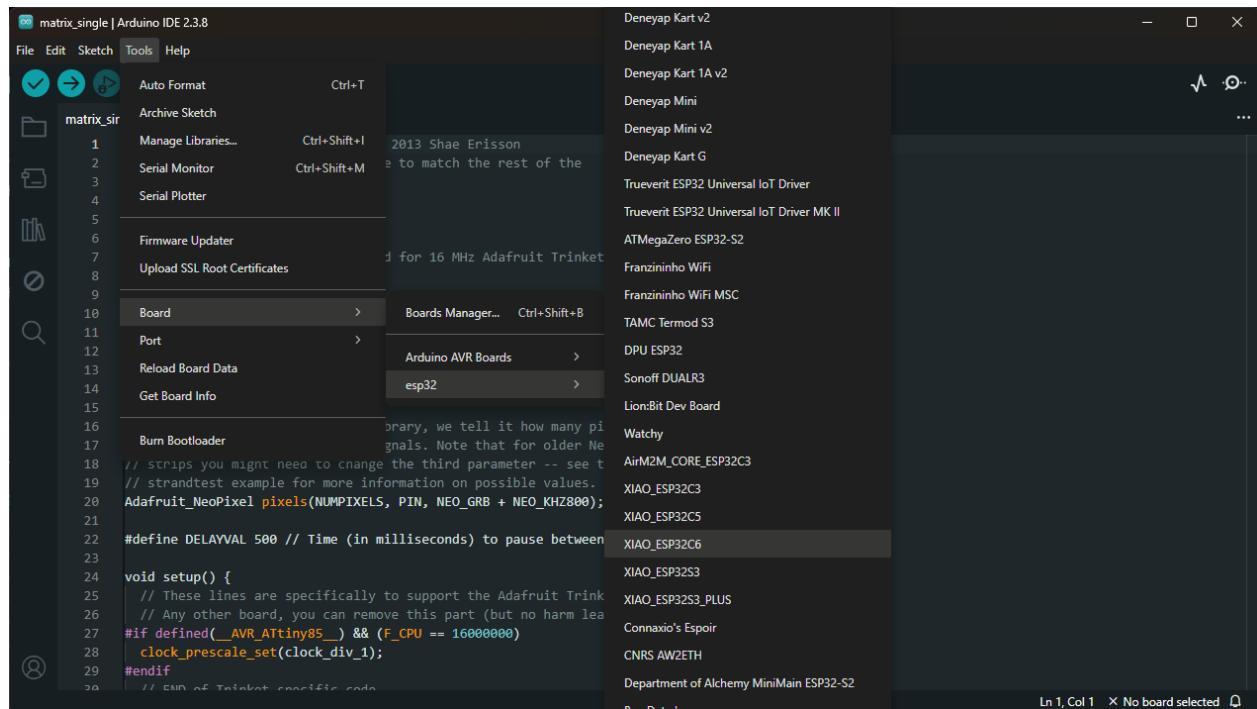
matrix_single.ino
1 // NeoPixel Ring simple sketch (c) 2013 Shae Erisson
2 // Released under the GPLv3 license to match the rest of the
3 // Adafruit NeoPixel library
4
5 #include <Adafruit_NeoPixel>
6 #ifdef __AVR__
7 #include <avr/power.h> // Required for 16 MHz Adafruit Trinket
8 #endif
9
10 // Which pin on the Arduino is connected to the NeoPixels?
11 #define PIN D10 // On Trinket or Gemma, suggest changing this to 1
12
13 // How many NeoPixels are attached to the Arduino?
14 #define NUMPIXELS 25 // Popular NeoPixel ring size
15
16 // When setting up the NeoPixel library, we tell it how many pixels,
17 // and which pin to use to send signals. Note that for older NeoPixel
18 // strips you might need to change the third parameter -- see the
19 // strandtest example for more information on possible values.
20 Adafruit_NeoPixel pixels(NUMPIXELS, PIN, NEO_GRB + NEO_KHZ800);
21
22 #define DELAYVAL 500 // Time (in milliseconds) to pause between pixels
23
24 void setup() {
25   // These lines are specifically to support the Adafruit Trinket 5V 16 MHz.
26   // Any other board, you can remove this part (but no harm leaving it):
27   #if defined(__AVR_ATtiny85__) && (F_CPU == 16000000)
28     clock_prescale_set(clock_div_1);
29   #endif
30   // END of Trinket specific code.
31 }
```

Step 4. Setup Board and the Port

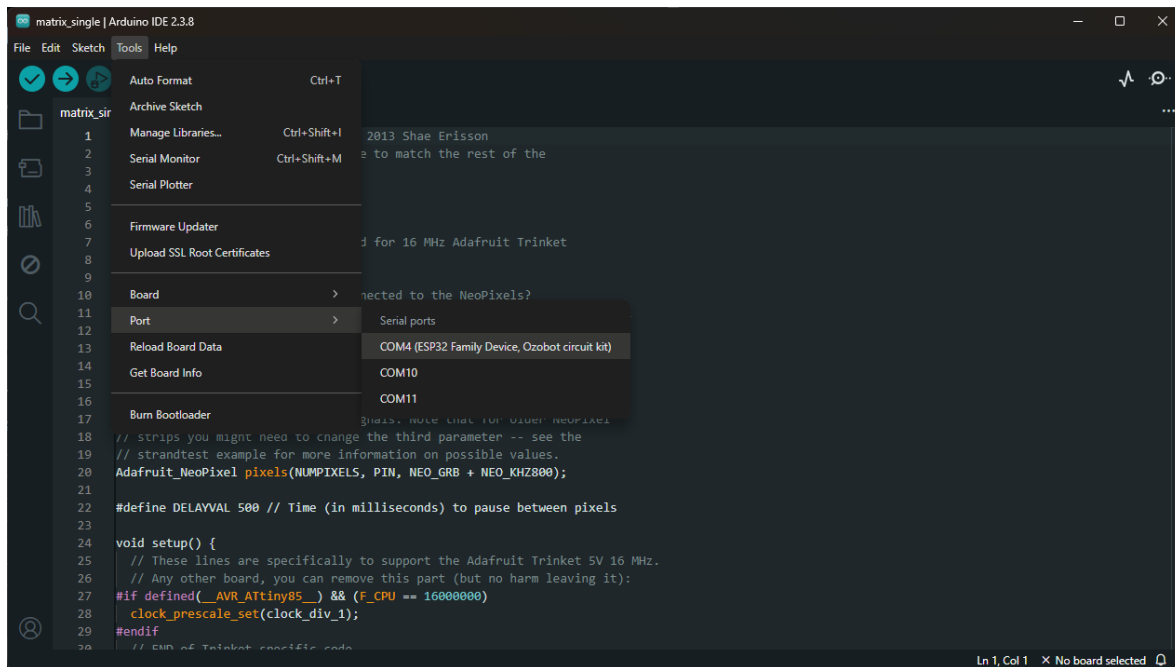
- Connect a USB-C cable from PC to the microcontroller board directly.



- Select the right board from **Tools -> Boards -> esp32 -> XIAO_ESP32C6**

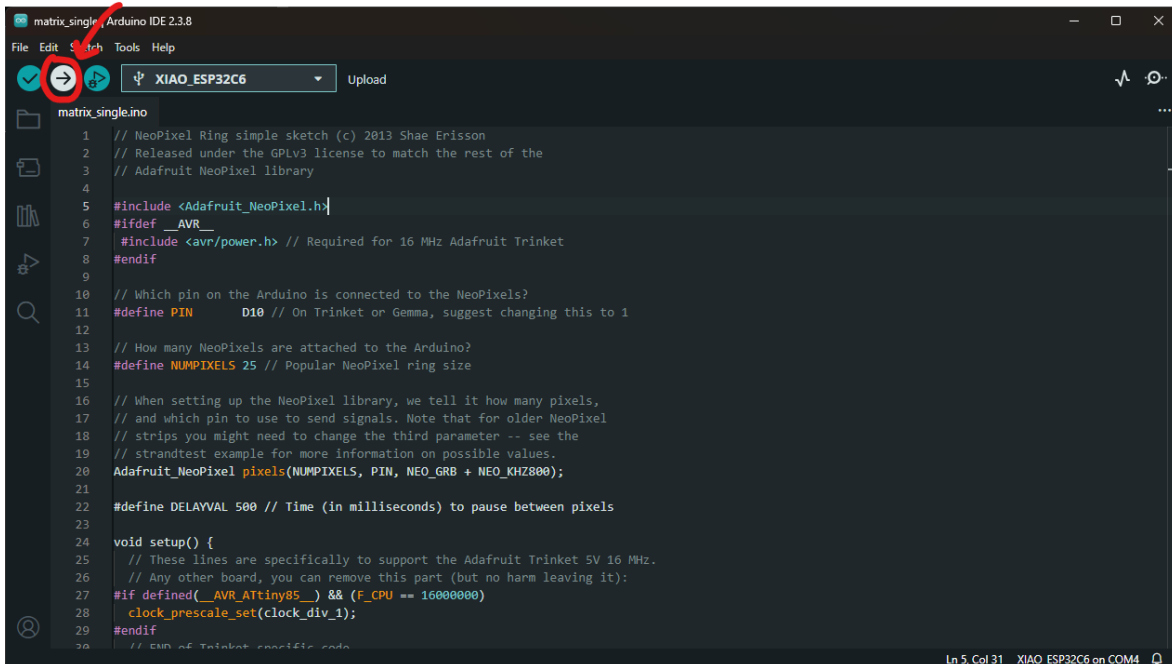


- Select the right port from **Tools -> Port -> COM3** (or higher). Choose the port that has the description “ESP32 Family Device....”

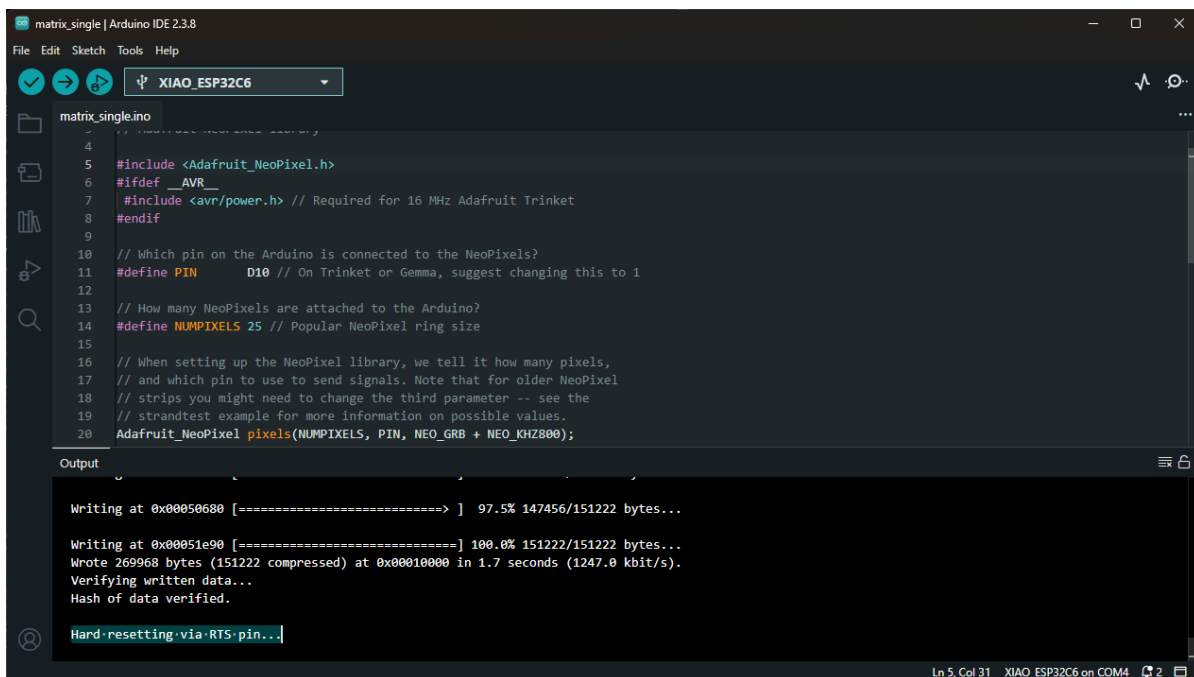


Step 5. Upload and Test

- Click Upload icon on top left.



- The program will compile and upload the compiled binary on the microcontroller board. It may take 5-10 min. Once uploaded the Output screen should print “Hard resetting via RTS pin...”



Step 6. Success

- Success! The lamp should light up!
- Unplug the USB